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## Lab Besos Work Flow from idf to surrogate desgin optimization

In this laboratory you will go over some of the basic work flow to create a a surrogate design optimization model from an EnergyPlus simulation. You will train a surrogate model network to find optimal design parameters.

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```
#import modules  
%matplotlib inline
```

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## Building presentation

In this Laboratory you will have to perform a surrogate optimization of a sample residential building unit.

The considered unit is composed by a typical flat of a multi-storey building. Its internal organisation and definition is in line with suggested residential building typologies included in well-known architectural technical manuals.

The considered building is composed by two units for floor, while a single unit is simulated for this chapter. The simulated spaces are considered to be at an intermediate floor with an upper floor and lower floor working at the same temperature (adiabatic). Similarly, the simulated unit is touching a specular one: confining walls are also assumed as adiabatic. Upper-floor balconies are included to consider shading effects.

Construction Parameters for the optimization:

- 1) Wall and Roof have each one a a proper insulation material
- 2) Windows can be chose between single, double or triple glazing
- 3) ACH ventilation can be set to a defined value
- 4) The watt/m2 of lights can be chosen for each zone
- 5) Windows shading set point can be chosen for both temperature and solar radiation



## Load the Building

The building is defined by the Information Data File (IDF) or using the new EnergyPlus format (IDF).

```
In [2]: # Open the IDF file lab_building.idf
```

## Evaluator

### Set up the inputs and outputs of your exploration

- what properties of the building will we be changing?
- what are some of the performance metrics of the building that we want to explore?
- what external weather conditions is the building experiencing?



### Load and Display Weather Data

```
In [3]: #The weather conditions are specified in the EnergyPlus Weather File (EWP
```

## Define The Problem

1. Define the parameters and your objectives you want to change
2. Create Selectors for getting the parameters fields in the model
3. Describe the parameter variation
4. Define your problem

```
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```

## Generate the Dataset

1. Generate 10 samples with sampling strategy
2. Setup the parallel processing
3. Simulate the Samples
4. Store and recover the expensive runs

```
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Run the samples and calculate the execution time

```
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Calculate the time

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## Store the expensive calculations

Since this can quite a big run. Lets store the results such that we don't have to rerun this problem.

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## Analize and describe your simulation output

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## Visualising the parametric analysis

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## Correlation heat map

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## Setup the dataset for the Surrogate Model

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Try with more samples 20 and 40

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```
#You can load the data from the files in the sim_samples_folder  
# 20 samples generation took: 5 min 18 seconds  
# 40 samples generation took: 19 min 7 seconds
```

## Surrogate Modelling Evaluator object

We can wrap the fitted model in a BESOS Evaluator .

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## Exploration

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# Perform Building Optimization

Using the best surrogate perform an optimization process, select optimal values and save a new idf with the selected values. Evaluate the goodness of the surrogate simulation with EnergyPlus.

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Identify best building design parameters and save the new idf

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